1. A particle carrying a charge of 0.5 μC enters a magnetic field of strength 0.045 T with a velocity of 350 m/s. The velocity is perpendicular to the magnetic field. What is the magnetic force acting on the charged particle?

2. A particle carrying a charge of 1.2 μC enters a magnetic field of strength 0.45 T with a velocity of 200 m/s. The velocity is 45° to the magnetic field. What is the magnetic force acting on the charged particle?

3. What magnetic field strength is needed to exert a force of 1.0×10^{-15} N on an electron traveling at 2.0×10^{7} m/s?

4. A 5 cm segment of wire carrying 2.5 A is perpendicular to a magnetic field of 25 T inside a solenoid. What is the magnetic force on the segment? If the current inside the solenoid is increased to 5 times its original value, what will the new magnetic field strength be?

5. A solenoid 0.20 m long has 600 turns of wire. What current must be passed through the solenoid to produce a magnetic field of 2.0×10^{-2} T.

6. A 61 mg mass just balances the balance arm of a current balance when the strip current is 3.0 A. If the strip is 2.2 cm long, what is the magnetic field strength inside the solenoid in which the current balance is located?